Chain of dependence

In *Animal Ecology*, Elton outlined the key principles of the study of animal communities: food chains and webs, food size, and ecological niches. Each food chain and web, he asserted, is dependent on producers: plants and algae that support plant-eating consumers (herbivores). These herbivores in turn support one or more levels of meat-eating consumers (carnivores). Large carnivores generally eat smaller animals, but because small animals reproduce more quickly, their numbers are able to support the larger predators.

Competition for food is very tight near the top of a food web. Although apex (top) predators, such as big cats and large birds of prey, have no natural predators, this often means that they have to defend territories against rivals of their own species to ensure there is enough food for themselves and their offspring.

“Every animal is closely linked with a number of other animals living around it—and these are largely food relations.”

Charles Elton